

setup. This can also be done in the same file under 'GPIO NAME' section (see Fig. 8).

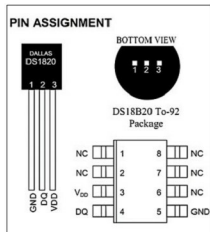


Fig. 11: DS18B20 pin configurations



Fig. 12: TV control

SSH connection terminal

You can start a terminal session on Raspberry Pi with your phone using SSH (see Fig. 9). For that, just press the symbol for 'Shell' (Fig. 1) and enter the IP of your Raspberry Pi. Also put the username and password. You will be logged in to Raspberry Pi through a remote session within seconds.

Temperature sensor monitoring

The system can monitor and graph temperatures remotely sensed by programmable resolution 1-wire digital thermometer DS18B20. It communicates over a 1-wire bus that by definition requires only one data line for communication with a central microprocessor. The DS18B20 digital thermometer provides 9-bit to 12-bit Celsius temperature measurements and has an alarm function with non-volatile user-programmable upper and lower trigger points. (Refer Fig. 11 for pin description.)

The moment you connect this sensor to GPIO4 (BCM GPIO numbering), it will reflect in the panel shown in Fig. 10 and the current temperature will be indicated.

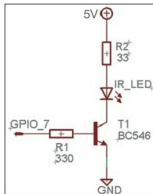


Fig. 13: IR circuit for remote control

TV control

You can easily control a TV with your Android phone acting as the remote (refer Fig. 12). This function can be implemented with a small circuit shown in Fig. 13. Connect this circuit to GPIO7 (BCM GPIO number) as shown in the circuit and control your TV.

Process monitoring

The process monitor panel shown in Fig. 14 can help you view and control various processes running on Raspberry Pi. You can stop/run any process that you like. The panel also shows the CPU and memory usage per process for convenience.

Last but not the least



Fig. 14: Process monitoring panel

The functionalities mentioned above are more than enough to implement a home automation system controlled by your Android phone. To add even more to it, there are functions like live camera feed and I²C interface also implemented. The settings are simple and can easily be configured. ●

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